

Christina Paolicelli

Embedded Software Engineer

CONTACT

Phone: 203-249-3712

Portfolio: www.hriste.github.io

Email: Christina.dot.paolicelli@gmail.com

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Algorithms | 2023 – Present

Shaper Tools – Handheld computer-vision-enabled CNC Router

- Designed and implemented computational geometry algorithms for automated pocket toolpath generation, including polygon offsetting, region decomposition, and robust handling of self-intersecting and nested contours.
- Improved high-speed cutting performance by optimizing toolpath generation strategies.
- Implemented computer vision-based marker detection for secure, unique pairing with a Bluetooth accessory.
- Modularized core geometry and vision systems into standalone libraries, enabling deterministic testing, improving reusability across embedded and WebAssembly targets.

Senior Software Engineer | 2021 – 2023

Rain Industries – Drones for Wildfire Response

- Lead software design and implementation for Fire Mapping. A system that identifies, locates, and suppresses wildfires from an autonomous helicopter.
- Stood up sensor test platform drone, coordinated test efforts, and wrote data analysis tooling.
- Operations crew member for autonomous helicopter test flight campaigns.

Senior Software Engineer | 2020 – 2021

Software Engineer | 2017 – 2020

BAE Systems

- Designed, developed, and tested software for embedded Electronics Warfare applications.
- Developed automated testing infrastructure using Python and VBA.
- Reverse engineered sensor systems signal acquisition processes.
- Designed and developed proof of concept motor control and communications software.

EDUCATION

Johns Hopkins University | Grad 2020

Master of Science in Computer and Electrical Engineering

Thesis: Deep Learning Framework for Character Recognition in Low Quality License Plate Images

Rensselaer Polytechnic Institute | Grad 2017

Bachelor of Science in Electrical Engineering

SKILLS

C++

Python

Git

INTERESTS

Figure Skating

Snowboarding

Knitting